

3. Core Python Concepts

➡ Understanding data types: integers, floats, strings, lists, tuples, dictionaries, sets.

- Value means data and it's type called data type
- Mainly two types data type.

1. Primitive

- Called basic data type.
- Represent single value.

➤ Type

- Integer (int)
- Floating (float)
- Boolean (bool)
- String(str)

2. Non-primitive

- Used to store collection of values or complex structure.

➤ Type

1. List (list)
2. Tuple(tuple)
3. Dictionaries(dict)

Python variables and memory allocation.

➤ Variable.

- A variable is a name that refers to a value stored in memory.
- “a=10” a is a variable.

➤ Memory allocation.

- When a variable is assigned a value, Python creates an object in memory and the variable refers (or points) to that object.
- if multiple variables refer to the same object, Python does not create a new object unless the value is changed.

Python operators: arithmetic, comparison, logical, bitwise.

- Python provides several types of operators. Here are four important ones:

1. Arithmetic Operators.

- Used for mathematical calculations.
- Addition a+b

- Subtraction $a - b$
- Multiplication $a * b$
- Division a / b

2. Comparison Operators

- Equal to $a == b$
- Not equal to $a != b$
- Greter than $a > b$
- Less than $a < b$
- Greter or equal $a >= b$
- Less or equal $a <= b$

3. Logical Operators

- And $(a > 5 \text{ and } b < 5)$
- Or $(a < 5 \text{ or } b < 5)$
- Not $\text{not}(a > b)$

4. Bitwise Operators

- $\&$ and
- $\mid$ or
- $\wedge$ xor
- $\sim$ not